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Studierichting: Informatica
Oefeningenreeks 4A nr. 14.8

$$I = \int x^2 \ln^2 x dx$$

$$\left\{ \begin{array}{rl} u = & \ln^2 x \\ dv = & x^2 dx \end{array} \right. \Rightarrow \left\{ \begin{array}{rl} du = & \dfrac{2\ln x}{x} dx \\ v = & \dfrac{x^3}{3} \end{array} \right.$$

$$I = \ln^2 x \cdot \frac{x^3}{3} - \int \frac{x^3}{3} \cdot \frac{2\ln x}{x} dx$$

$$= \frac{1}{3} x^3 \ln^2 x - \frac{2}{3} \int x^2 \ln x dx$$

$$\left\{ \begin{array}{rl} u = & \ln x \\ dv = & x^2 dx \end{array} \right. \Rightarrow \left\{ \begin{array}{rl} du = & \dfrac{1}{x} dx \\ v = & \dfrac{x^3}{3} \end{array} \right.$$

$$I = \frac{1}{3} x^3 \ln^2 x - \frac{2}{3} \left(\frac{1}{3} x^3 \ln x - \int \frac{x^3}{3} \cdot \frac{1}{x} dx \right)$$

$$= \frac{1}{3} x^3 \ln^2 x - \frac{2}{3} \left(\frac{1}{3} x^3 \ln x - \frac{1}{3} \int x^2 dx \right)$$

$$= \frac{1}{3} x^3 \ln^2 x - \frac{2}{3} \left(\frac{1}{3} x^3 \ln x - \frac{1}{3} \cdot \frac{x^3}{3} \right)$$

$$= \frac{1}{3} x^3 \ln^2 x - \frac{2}{9} x^3 \ln x + \frac{2}{27} x^3 + c$$

References